Assignment 4 – Ethical AI Questions

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Introduction

Artificial intelligence now makes or shapes decisions that used to belong to people: who gets interviewed, what a car does in an emergency, where police patrol, and how a network allocates its resources. Ethical AI is the discipline of making sure those systems serve human values rather than quietly working against them. This assignment answers thirty questions in three parts: conceptual foundations and the major international frameworks (IEEE, UNESCO, OECD, and the EU AI Act), applied scenarios where ethical principles are tested in practice, and reflective questions about where AI ethics is heading and my own role in it. Where it is relevant, I connect the answers to what I built earlier in this course, because the intelligent, self-optimizing networks we studied raise exactly these questions.

Section A: Conceptual Understanding

Question 1 — What is Ethical AI, and why is it essential in modern technological systems?

Ethical AI is the practice of designing, building, and operating AI systems so that they respect human rights, treat people fairly, protect privacy, and remain under meaningful human control. It is essential because AI systems now make high-stakes decisions at a scale and speed no human process can match: a biased hiring tool does not discriminate against one applicant, it discriminates against every applicant, consistently and invisibly. Ethics is what keeps automation aligned with the values of the people it affects, and it is also practical — systems that are fair, transparent, and accountable are the ones people and regulators will actually trust and adopt.

Question 2 — List and explain any four major ethical principles that guide AI development.

Four widely shared principles are: **fairness and non-discrimination** — the system should not produce systematically worse outcomes for people based on characteristics such as gender, race, or age; **transparency** — those affected should know that AI is being used and understand, at an appropriate level, how it works; **accountability** — identifiable people and organizations must remain answerable for what the system does, with routes for appeal and redress; and **privacy** — the system should collect only the data it needs, protect it, and use it only for the stated purpose. These four appear, in some form, in essentially every major framework, including the OECD principles and UNESCO's Recommendation (OECD, 2019; UNESCO, 2021).

Question 3 — What is the main objective of the IEEE Ethically Aligned Design framework?

The main objective of IEEE's *Ethically Aligned Design* is to ensure that autonomous and intelligent systems are designed to prioritize human well-being — not just performance or profit — and that ethical considerations are built in from the start of the design process rather than patched on afterward (IEEE, 2019). It translates broad values such as human rights, data agency, transparency, and accountability into concrete guidance that engineers and organizations can apply while they are building the system, which is when ethical choices are still cheap to make.

Question 4 — Define the term algorithmic bias and provide one real-world example.

Algorithmic bias is a systematic tendency of an AI system to produce unfair outcomes for particular groups, usually because the data it learned from reflects historical inequities or unrepresentative sampling, or because of choices made in the system's design. A well-documented example is the experimental recruiting tool at Amazon that learned from a decade of past hiring data — mostly male applicants — and began penalizing resumes that included the word "women's," effectively downgrading female candidates; the project was scrapped (Dastin, 2018).

Question 5 — Explain the difference between transparency and explainability in AI.

Transparency is openness about the system itself: disclosing that AI is being used, what data it was trained on, what it is for, and what its known limitations are. Explainability is narrower and decision-focused: the ability to give a person an understandable reason for a *specific* output — why this loan was denied, why this resume was ranked low. A company can be transparent (publishing model documentation) while its model remains unexplainable (a black box that cannot justify individual decisions), and vice versa. Mature governance requires both: transparency builds systemic trust, explainability makes individual recourse possible.

Question 6 — What role does accountability play in the ethical use of AI systems?

Accountability is what prevents "the algorithm decided" from becoming an excuse. It means a specific person or organization is answerable for an AI system's outcomes, that decisions can be audited and traced, and that harmed individuals have a route to challenge them and obtain redress. Accountability also changes incentives before deployment: teams that know they will answer for a system's failures test it more carefully. Without accountability, every other principle is voluntary; with it, fairness, transparency, and safety have an enforcement mechanism.

Question 7 — Identify three international organizations involved in developing AI ethics guidelines.

Three prominent ones are **UNESCO**, whose Recommendation on the Ethics of Artificial Intelligence was adopted by its member states in 2021; the **OECD**, whose AI Principles (2019) were the first intergovernmental AI standard; and the **IEEE**, whose Ethically Aligned Design initiative and related standards translate ethics into engineering practice (IEEE, 2019; OECD, 2019; UNESCO, 2021). Others include the European Union, ISO/IEC, and the ITU.

Question 8 — What does UNESCO's Recommendation on the Ethics of Artificial Intelligence emphasize?

UNESCO's Recommendation — the first global standard on AI ethics, adopted by 193 member states — grounds everything in human rights and human dignity. It emphasizes proportionality and doing no harm, fairness and non-discrimination, privacy, human oversight (ultimate responsibility must always rest with humans, not machines), transparency and explainability, accountability, and sustainability of both society and the environment (UNESCO, 2021). It is notable for going beyond principles into policy action areas — data governance, education, labor, gender, and environment — and for opposing uses it considers incompatible with human rights, such as social scoring and mass surveillance.

Question 9 — What are OECD's five key principles for trustworthy AI?

The OECD's five values-based principles are: (1) AI should benefit people and the planet through **inclusive growth, sustainable development and well-being**; (2) AI should respect **human rights, democratic values, and fairness**, including privacy and equality; (3) there should be **transparency and explainability**, so people can understand and challenge AI outcomes; (4) AI systems must be **robust, secure and safe** throughout their lifecycle; and (5) organizations and individuals deploying AI must be **accountable** for its proper functioning (OECD, 2019). Forty-plus countries have adhered to them, making them the most widely adopted intergovernmental baseline.

Question 10 — How does the EU AI Act classify and regulate AI systems based on risk?

The EU AI Act (Regulation (EU) 2024/1689) takes a risk-tiered approach. **Unacceptable-risk** practices — such as social scoring by public authorities, manipulative techniques that cause harm, and certain forms of untargeted biometric surveillance — are banned outright. **High-risk** systems — for example AI used in hiring, credit scoring, critical infrastructure, education, and law enforcement — are permitted but heavily regulated: they require risk management, high-quality training data, logging, human oversight, and conformity assessment before market entry. **Limited-risk** systems carry transparency duties, such as telling users they are talking to a chatbot and labeling AI-generated content. **Minimal-risk** systems (the vast majority) face no new obligations. The Act also adds specific obligations for general-purpose AI models (European Union, 2024).

Section B: Applied and Scenario-Based

Question 11 — An AI recruitment tool consistently prefers male candidates. Which ethical principles are violated?

The primary violation is **fairness and non-discrimination**: the system produces systematically worse outcomes for women, which in employment is also illegal discrimination in most jurisdictions. **Transparency and explainability** are typically violated too, because candidates are unaware an algorithm screened them out and are given no reason. If the organization keeps using the tool without auditing it, **accountability** is violated as well — no one is answering for a known harm. The Amazon case shows the root cause is usually not intent but training data that encodes a biased history (Dastin, 2018), which is precisely why proactive bias testing is an ethical obligation, not an optional extra.

Question 12 — A self-driving car must choose between hitting one pedestrian or five. Which ethical dilemma does this represent?

This is the **trolley problem**, the classic moral dilemma that pits utilitarian reasoning (minimize total harm — sacrifice one to save five) against deontological reasoning (there is a moral difference between allowing harm and actively choosing to inflict it). Embedded in a vehicle, it becomes real engineering: someone must decide, in advance and in code, whose safety is weighted how. The Moral Machine experiment, which collected almost forty million such judgments worldwide, showed that people's moral preferences differ across cultures — which means there is no single "correct" setting to program, and the choice itself demands transparency and public accountability (Awad et al., 2018).

Question 13 — In predictive policing, how can bias be introduced into the algorithm's training data?

Predictive policing models are typically trained on historical crime records, and those records measure *enforcement*, not crime. If certain neighborhoods were over-patrolled in the past, they generate more recorded incidents, the model flags them as high-risk, more patrols are sent, more incidents are recorded, and the loop reinforces itself — a feedback loop that launders past practice into "objective" prediction. Bias also enters through label choice (arrests are not convictions), through under-reporting of crimes in other areas, and through proxy variables like zip code that correlate with race. The result can be discriminatory policing with a statistical veneer.

Question 14 — How might a "human-in-the-loop" system reduce ethical risks in automated decision-making?

A human-in-the-loop design keeps a person between the model's output and the consequential action: the AI recommends, a human decides. This reduces risk in three ways. It catches errors and context the model cannot see — an unusual case, a data glitch, a circumstance the training data never contained. It preserves a clear point of accountability, because a named person approved the action. And it slows automation exactly where stakes are highest, which is what the EU AI Act's human-oversight requirement for high-risk systems formalizes (European Union, 2024). The caveat is that the human must be genuinely empowered and attentive; a rubber-stamp reviewer provides the appearance of oversight without the substance.

Question 15 — Describe how data privacy laws such as GDPR affect the design of AI systems.

GDPR shapes AI design from the first line of the data pipeline. It requires a lawful basis and a stated purpose for processing personal data, which restricts opportunistic data hoarding; **data minimization**, which forces teams to justify every feature they collect; and **privacy by design and by default** as an architectural obligation. Individuals gain rights that the system must be built to honor — access, correction, and erasure, which is nontrivial for trained models. Most importantly for AI, Article 22 gives people the right not to be subject to solely automated decisions with legal or similarly significant effects, and a right to meaningful information about the logic involved — pushing systems toward human review and explainability (European Union, 2016).

Question 16 — A generative AI creates a fake video of a politician. What ethical issues arise here?

Several at once. **Deception and manipulation**: the video can mislead voters and distort democratic processes. **Consent and dignity**: the politician's likeness is used to say things they never said. **Erosion of trust**: as deepfakes spread, real recordings become deniable too — the "liar's dividend," where genuine evidence can be dismissed as fake. There are also downstream harms: defamation, incitement, and market or diplomatic consequences before a correction catches up. This is why transparency obligations — labeling synthetic media, provenance watermarking — appear in the EU AI Act, and why platforms and toolmakers share responsibility with the individual who made the video (European Union, 2024).

Question 17 — What are the ethical implications of AI-powered surveillance systems in public spaces?

Mass AI surveillance inverts the default of public life: instead of being anonymous unless suspected, everyone is tracked by default. That erodes **privacy**, chills freedom of expression and assembly (people behave differently when watched), and concentrates power in whoever controls the system. The technical failures compound the ethical ones: facial recognition has shown markedly higher error rates for darker-skinned women than for lighter-skinned men, so the burden of misidentification falls unevenly (Buolamwini & Gebru, 2018). There is also function creep — systems installed for one purpose quietly expanded to others. These concerns are why UNESCO opposes mass surveillance uses and why the EU AI Act tightly restricts real-time remote biometric identification in public (European Union, 2024; UNESCO, 2021).

Question 18 — How can organizations ensure transparency in AI-driven financial decisions?

Practically: disclose that an algorithm is involved in the decision; provide specific, understandable reasons with every adverse outcome (the "reason codes" behind a credit denial); prefer interpretable models where stakes are high, or pair complex models with reliable explanation methods; document models formally — purpose, data, performance across demographic groups, limitations; run independent audits and fair-lending analyses; and offer an appeal channel that reaches a human with the authority to overturn the decision. Financial services also sit in the EU AI Act's high-risk category, so much of this is becoming law rather than good practice (European Union, 2024).

Question 19 — Discuss the ethical concerns surrounding AI in healthcare diagnostics.

The stakes make every principle sharper. **Bias**: a diagnostic model trained mostly on one population can be systematically less accurate for others — for example, skin-lesion models trained largely on lighter skin. **Accountability**: when an AI-assisted diagnosis is wrong, responsibility must be clearly assigned among clinician, hospital, and vendor, or patients are left without recourse. **Explainability**: clinicians should not act on predictions they cannot interrogate. **Privacy**: health data is the most sensitive category there is. And **automation bias**: clinicians may defer to the machine even when their own judgment disagrees, or gradually lose skills the AI has absorbed. The ethical consensus is that diagnostic AI should function as decision support under meaningful human oversight, validated clinically like any medical device — not as an autonomous diagnostician.

Question 20 — What measures can be implemented to make AI datasets more representative and fair?

Start by measuring: audit the dataset's demographic composition and test model performance disaggregated by group, because an average accuracy number hides unequal error rates (Buolamwini & Gebru, 2018). Then act on what the audit shows: source data deliberately from under-represented groups; rebalance or reweight classes; document every dataset formally (what it contains, how it was collected, what it should not be used for); involve diverse annotators and review labeling guidelines for embedded assumptions; and keep monitoring after deployment, because populations and data drift. Fairness is not a one-time preprocessing step — it is a maintenance discipline, the same way I learned in this course that model evaluation is.

Section C: Reflective and Future-Oriented

Question 21 — Do you think AI can ever be completely unbiased? Justify your answer.

No — and I think the honest framing matters. Models learn from data produced by an imperfect world, so some bias always rides in with the signal; even a perfectly clean dataset cannot resolve the deeper problem that "unbiased" itself requires normative choices. Research on fairness has shown that reasonable mathematical definitions of fairness can be mutually incompatible — satisfying one can force violating another — so someone must decide which fairness matters most for this system, and that decision is ethical, not technical. The realistic goal is therefore not zero bias but *managed* bias: measured, disclosed, minimized, and monitored. My time-series work in this course taught me the small version of this lesson — there is no perfectly honest model, only models whose assumptions and leaks you have found and dealt with.

Question 22 — How can ethical frameworks ensure responsible innovation in AI?

Frameworks work when they convert values into engineering requirements. IEEE's Ethically Aligned Design turns "well-being" into design guidance; the OECD principles give governments and companies a shared vocabulary; the EU AI Act attaches obligations and penalties to risk levels (European Union, 2024; IEEE, 2019; OECD, 2019). Their real function is to move ethics upstream: a bias audit at design time costs little, while a discrimination scandal after deployment costs trust that may never return. Frameworks also level the competitive field — when every player must meet the same floor, no one gains an advantage by cutting ethical corners. They cannot *ensure* responsibility on their own; they need audits, enforcement, and a professional culture that treats the checklist as a starting point rather than the finish line.

Question 23 — Should governments enforce stricter AI regulations, or should industries self-regulate? Discuss.

I think the evidence points to a hybrid, tiered the way the EU AI Act does it. Pure self-regulation fails under competitive pressure — a company that voluntarily slows down for safety watches its competitors ship; the incentives guarantee that someone defects. Pure government control fails differently: legislation moves in years while the technology moves in months, and overly broad rules can burden harmless applications. The workable division is government setting a binding floor for high-stakes uses — bans on unacceptable practices, mandatory oversight and audits for high-risk systems, real penalties — while industry builds the fast-moving technical standards (testing methods, documentation formats, red-teaming practice) that regulators then reference. Self-regulation works best as the implementation layer of regulation, not as a substitute for it.

Question 24 — What ethical challenges might Quantum AI pose in the future?

Two stand out. First, **security and privacy**: a sufficiently capable quantum computer breaks the public-key encryption protecting today's communications — and adversaries are already harvesting encrypted data to decrypt later. Since everything I studied in this course runs on encrypted networks, "quantum-safe" migration is an ethical obligation to users, not just a technical upgrade. Second, **concentration of power**: quantum hardware is extraordinarily expensive, so quantum-accelerated AI would initially belong to a handful of states and corporations, widening the capability gap that already worries AI-ethics bodies. Add auditability — quantum-enhanced models could be even harder to inspect than today's networks — and the lesson is that governance needs to arrive *before* the capability, for once.

Question 25 — How might Generative AI reshape intellectual property rights?

It is straining the system at both ends. On the input side: models are trained on billions of works whose creators never consented and are not compensated — the wave of ongoing litigation will decide whether that is fair use or infringement, and licensing markets for training data are already emerging in anticipation. On the output side: most copyright regimes require human authorship, so purely AI-generated works may be unprotectable, while "how much human contribution is enough" becomes the new boundary question. In between sit style imitation — legal in letter, corrosive in spirit when a living artist's market is undercut by their own aesthetic — and provenance: watermarking and content credentials so audiences know what they are looking at. I expect disclosure duties and collective licensing to become as normal for AI as sampling clearances became for music.

Question 26 — What are the moral responsibilities of AI developers toward end-users?

Developers hold knowledge and power that users do not, and responsibility follows that asymmetry. Concretely: anticipate harm and misuse before release, not after the incident; test for bias and safety and publish honest limitations rather than marketing claims; protect user data as a duty, not a feature; design so users know when they are dealing with AI and can reach a human when it matters; provide recourse when the system is wrong; and be willing to say no to applications that cannot be built responsibly. Other engineering professions accepted long ago that the public's safety outranks the client's preference; software affecting millions deserves the same professional ethic.

Question 27 — How can AI sustainability contribute to global ethical goals?

In both directions. AI's own footprint is an ethical issue — training and serving large models consumes serious energy and water, so efficiency is a moral choice, not just a cost one. My edge-computing lab made this tangible: the distilled model was ten times smaller than the baseline with modest accuracy loss, and at the scale of millions of edge devices that difference is measured in megawatts. In the other direction, AI is a genuine sustainability tool: the O-RAN energy-saving applications we studied let networks power down radio resources intelligently, and similar optimization applies to grids, buildings, and logistics. Both directions align with the sustainability emphasis in UNESCO's Recommendation and the OECD's "benefit people and planet" principle (OECD, 2019; UNESCO, 2021) — efficient AI serving efficiency is ethics and engineering pointing the same way.

Question 28 — What future jobs might emerge to address ethical issues in AI systems?

A whole compliance-and-assurance profession is forming, and much of it maps onto roles that already exist in IT governance. I expect: **AI auditors** who independently test systems for bias, robustness, and regulatory conformity, the way financial auditors certify accounts; **responsible-AI officers** owning ethics policy at the executive level; **AI compliance specialists** managing conformity assessments under the EU AI Act, a direct analog of GDPR data-protection officers; **red-team specialists** paid to make models fail before adversaries do; **model documentation and provenance specialists**; and **AI incident responders** who investigate failures with the discipline my ITSM field applies to outages — timeline, root cause, corrective action. Coming from IT service management, I find this trajectory familiar: every technology that became critical infrastructure eventually grew a profession dedicated to keeping it trustworthy.

Question 29 — In your opinion, what will be the most pressing AI ethics issue in 2035, and why?

Accountability for autonomous AI agents. The trajectory is visible in this course: the networks we studied already predict their own traffic, reallocate their own resources, and heal their own failures, and the same agentic pattern is spreading into commerce, software, and infrastructure. By 2035 the pressing question will not be "is the model biased" — we will have tooling for that — but "who answers when a chain of autonomous decisions, made by interacting agents at machine speed with no human in the loop, causes real harm?" Our accountability concepts assume a human decision to point to; agent autonomy dissolves that anchor. Getting it back will require decision logging, liability regimes for delegated authority, and hard limits on what may be automated without oversight — and I would rather see that framework built before the first systemic agent-caused failure than after it.

Question 30 — Reflect on your role as a future professional — how would you ensure ethical practices in AI development?

My path runs through operating AI systems, not just building them, and operations is where ethical practice lives or dies. Concretely, I would carry four habits forward. First, honest evaluation: this course taught me, through a target-leakage bug that produced a perfect and worthless R^2 of 1.0, that a too-good metric is a bug report — I will hold AI systems to evidence, not demos. Second, ITSM discipline applied to models: change management before deployment, monitoring for drift and disaggregated error rates in production, and blameless-but-thorough incident review when an AI system misbehaves. Third, insistence on human oversight and appeal routes wherever a system touches people's opportunities. Fourth, documentation as an ethical act — my whole portfolio this term was an exercise in making work transparent and reproducible, and that is exactly what accountable AI requires at professional scale. Ethics in AI will not be ensured by principles on a wall; it will be ensured by people who run the systems treating fairness, transparency, and accountability as operational requirements — and that is the professional I intend to be.

Conclusion

Across these thirty questions, one structure repeats. The foundations (Section A) show remarkable global agreement: fairness, transparency, accountability, privacy, human oversight, and sustainability appear in every major framework, from IEEE's design guidance to UNESCO's global standard to the OECD principles to the EU's binding, risk-tiered law. The scenarios (Section B) show why the agreement matters: without deliberate effort, hiring tools inherit historical bias, policing models launder over-enforcement into prediction, and surveillance quietly rewrites the defaults of public life. The reflections (Section C) show where the frontier is moving: from auditing individual models toward governing autonomous agents, sustainable AI, and the professional cultures that operate these systems. My takeaway from this course is that the intelligent network and the ethical network are the same engineering problem — both are only as trustworthy as the evaluation, oversight, and accountability built around them.

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